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Project Title:	Protection from Cosmic Radiation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Ground	**Solicitation / Funding Source:**	2022 Space Biology NNH22ZDA001N-SBBLEO E.11: Research Pathfinder
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
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Task Description:	Significant DNA damage from ionizing radiation is a major medical challenge for human exploration of the solar system. The Artemis missions emphasize the importance of exploring the effects of simulated deep space radiation. The nematode C. elegans is an ideal biologic model for these studies for many reasons. C. elegans has forgiving and simple culture requirements; it has shown robust growth in previous space flight experiments; the entire genome has been sequenced; and tools for genomic manipulation are freely available. Furthermore, the molecular machinery for repair of radiation damage in C. elegans is well defined. DNA damage leads to a wave of the repair proteins RPA-1, followed by RAD-51, that repair double stranded breaks. One can generate a biologic detector of DNA damage by deleting the RPA-1 gene so that the DNA damage accumulates without intervening repair. Hence, our first aim is to study the RPA-1 and RAD-51 radiation repair system in C. elegans after exposure to cosmic radiation. Histone gamma-H2AX has been a popular molecule to monitor and quantitate radiation induced foci following radiation damage. However, C. elegans lacks gamma-H2AX or a homolog. Homologous recombination is not the only metabolic pathway important in radiation protection. The broad family of intrinsically disordered proteins (IDPs) have several potential roles, termed molecular shield functions, including protection from radiation. Like tardigrades, rotifers, and some arthropods, nematodes express molecular shield proteins. The C. elegans late embryogenesis abundant (LEA) proteins are a biologically important, intrinsically disordered protein molecular shield whose active protein motif is a single 44 amino acid sequence. LEA proteins inserted into E.coli protect from radiation damage, showing LEA is sufficient to induce radiation protection. Our second aim is to study whether LEA can protect C. elegans DNA from damage by simulated cosmic radiation. Newer methods offer unprecedented resolution to the identification and quantification of radiation damage. Surface-enhanced Raman spectroscopy (SERS) is a non-destructive chemical analysis technique that can detect and identify diverse forms of DNA damage. Raman spectroscopy can be performed in situ in intact tissues, or on isolated DNA samples. Options for ground-based controls have improved dramatically. Recent improvements in laser handling equipment at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory have resulted in a simple and practical five-beam simulation of cosmic radiation. Ground-based studies are more accessible and relatively inexpensive compared to flight studies. Ground controls can include rotating suspension conditions to simulate effects of microgravity, which are known to attenuate the DNA repair process. Hypothesis and Specific Aims Hypothesis: The C. elegans intrinsically disordered protein molecular shield, late embryogenesis abundant (LEA) protein, provides protection from simulated cosmic radiation damage. Specific Aim 1: To characterize simulated cosmic radiation damage to C. elegans DNA. Specific Aim 2: To test whether the intrinsically disordered protein molecular shield, late embryogenesis abundant (LEA) protein, protects from simulated cosmic radiation damage.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The two complementary approaches we are using to characterize DNA damage will increase our understanding of the biologic responses and impacts of deep space environment on the physiology of an actively growing multicellular organism as a model of human physiology. LEA, as a molecular shield that protects against radiation damage, can lead to the development of biologics, including gene therapies that can be used in diverse organisms. Scientific return will be expanded by adding these samples to the NASA GeneLab Data System and the NASA Ames Life Sciences Data Archive for access by the research community. These studies will advance NASA technical and operational experience and lessons learned for the design and implementation of experiments using animal models for future Artemis platforms.
Task Progress:	Hypothesis: The C. elegans intrinsically disordered protein molecular shield, late embryogenesis abundant (LEA) protein, provides protection from cosmic radiation damage. Specific Aim 1: To characterize and quantify cosmic radiation damage to C. elegans DNA using: A) Enumeration of radiation-induced foci of GFP-linked RPA-1 B) Identify and quantification of DNA damage by Surface Enhanced Raman Spectroscopy (SERS) • Subcontract with University of Iowa, where the proposed strains are being cloned, has been negotiated and executed. o Graduate students have been recruited to work on the project. o The proposed strains are in the process of being cloned and crossed into a harmonized background strain. • Subcontract with Mass General Brigham / Harvard Medical School, where the Raman spectroscopy analysis is to be performed, has been negotiated and executed. o Nematode culture has been established locally in the lab in Boston, establishing free access to specimens for background control studies for the Raman spectroscopy analysis. • Contract with Bionetics, where the new flight hardware is to undergo final design and be built, has been negotiated and executed. o Flight hardware design is approaching readiness for initial design review. Exact science requirements have been discussed and reviewed in Zoom meeting, telecons, and e-mails. • All stakeholders interact and communicate freely: o As least weekly e-mails, phone calls, and Zoom. • SERS nanoparticles (gold nanostar, gold nanorods, and spheres) critical for SERS-based detection of DNA damage have been prepared and quality control-validated by their characterization by microscopy (Transmission Electron Microscopy/TEM, ultraviolet/UV). • Experiments have begun focusing on the radiation mimics bleomycin exposure and radiation exposure studies using the Fluorinated Ethylene Propylene (FEP) bags integral to the new spaceflight hardware, and spinpods to model microgravity conditions. Specific Aim 2: To test whether the intrinsically disordered protein molecular shield, late embryogenesis abundant (LEA) protein, protects from cosmic radiation damage. • Both the LEA knockout nematode strain and the LEA overexpression strain are in hand, as they have become available in the two years since grant submission. Bob Goldstein (at University of North Carolina - Chapel Hill), and Jon Hibshman (at Southern Methodist University), identified and cloned all the isoforms at LEA, facilitating cloning strains with complete knockout of all LEA isoforms. • The LEA knockout strain has been crossed into a harmonized background strain; then the relevant fluorescent markers

	to see radiation induced foci will be crossed in. • RAD-51-GFP has been successfully inserted into both wildtype and LEA1 knockout nematodes. RAD-51 is part of the radiation induced foci (RIF) complex we proposed to assay. • We compared radiation induced focus formation in wildtype N2 nematodes, compared to LEA1 knockou, during stages of meiosis to validate our RIF fluorescent assay, and test if LEA1 knockout induced more RIF formation during meiosis; cells deliberately form numerous DNA double-strand breaks (DSBs) in order to initiate homologous recombination, which exchanges genetic information between homologous chromosomes and promotes accurate chromosome segregation. The data shows that our assay has the sensitivity and specificity to identify and quantity RIFs. Second, LEA1 knockout greatly and significantly increases the numbers of RIFs present. The assay appears to have the dynamic range to detect DNA double breaks in our planned cosmic radiation simulation experiments. New Flight Hardware. • The new flight hardware design is complete, and preliminary design review completed. Having passed preliminary design review, the new spaceflight hardware components are being procured, or manufactured, and assembly initiated.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Hammond TG, Smolikove S, Panikkanvalappil S, Wells H, and Birdsall HH. "Protection from cosmic radiation: An Artemis II based initiative." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024